# CUSTOMER SEGMENTATION DASHBOARD

Project Explanation Report

**Project:** Customer Segmentation Analysis

**Notebook:** Customer_Segmentation_Analysis.ipynb

**Dashboard:** Customer_Segmentation_Dashboard.html

**Tools:** Python, Pandas, NumPy, Scikit-learn, Plotly, HTML

**Method:** RFM Analysis + K-Means Clustering

**Dataset:** Customer and transaction data

**Dashboard Type:** Single-page HTML dashboard

## 1. Introduction

The Customer Segmentation Dashboard is a data analytics and visualization project developed using Python and an HTML/Plotly dashboard. The main purpose is to analyze customer purchasing behaviour and convert transaction-level data into meaningful customer groups. The dashboard helps users understand customer value, activity, revenue contribution, and segment-level behaviour.

## 2. Problem Statement

Businesses have customers with different purchasing patterns. Treating every customer in the same way can reduce the effectiveness of marketing and retention activities. Manual analysis of transaction records can also be difficult and time-consuming. This project addresses the problem by grouping customers according to their purchasing behaviour and presenting the results through a simple dashboard.

## 3. Objectives

The objectives are to analyze customer purchasing behaviour, calculate Recency, Frequency, and Monetary values, group customers with similar behaviour, apply K-Means clustering, interpret the resulting groups using meaningful business labels, create a clear HTML dashboard, and provide useful business recommendations.

## 4. Dataset Description

The HTML dashboard reports a project dataset containing 1,200 customers and 11,348 transactions. The project folder contains customer and transaction CSV files, cleaned datasets, an RFM table, clustered RFM results, final segment results, and a master table. Customer-level RFM features are used as the basis for segmentation.

## 5. Tools and Technologies Used

Python was used for data analysis and preprocessing. Pandas supports data loading, cleaning, grouping, and aggregation. NumPy supports numerical operations. Scikit-learn is used for K-Means clustering. Plotly is used for dashboard visualizations, and HTML/CSS is used to present the final dashboard in a web browser. Jupyter Notebook is used to document the analysis.

## 6. Data Cleaning and Preprocessing

The project workflow includes loading the customer and transaction data, checking and preparing the data, creating customer-level features, and preparing the RFM measures for clustering. The transaction records are transformed into a customer-level analytical table before applying segmentation.

## 7. Exploratory Data Analysis

The analysis focuses on customer activity and value rather than only overall sales. Important measures include customer count, revenue, average order value, RFM behaviour, segment distribution, revenue by segment, and the relationship between customer recency and monetary value.

## 8. Key Performance Indicators

The dashboard contains KPI cards for Total Customers, Total Revenue, Average Order Value, and Champions Share. The dashboard reports 1,200 total customers, ■19,919,415 total revenue, ■1,755 average order value, and a 25.2% Champions share.

## 9. RFM Analysis

RFM analysis summarizes customer behaviour using three measures. Recency indicates how recently a customer purchased; lower recency generally means more recent activity. Frequency represents how often the customer purchases. Monetary represents the total amount spent by the customer. These measures provide a simple and useful foundation for customer segmentation.

## 10. K-Means Clustering

K-Means is an unsupervised machine-learning algorithm used to group customers with similar RFM characteristics. Customers are assigned to clusters according to their similarity to cluster centroids. The resulting clusters are then interpreted in business terms instead of being left as numerical cluster IDs.

## 11. Dashboard Features

The project uses a single-page HTML dashboard titled “Customer Segmentation Analysis Dashboard”. It contains KPI cards, customer-segment explanation cards, a segment overview, customer distribution by segment, revenue by segment, and a customer-level Recency versus Monetary visualization. Plotly is used for the visual charts.

## 12. Customer Segments

The dashboard presents four business-oriented customer groups. Champions are recent, frequent, high-spend customers and should be prioritized for retention and loyalty rewards. At Risk customers were previously valuable but are becoming inactive and should receive re-engagement campaigns. Price-Sensitive / Low-Value customers make frequent smaller purchases and provide upselling or bundle opportunities. Dormant / Lost customers have been inactive for a longer period and may be targeted with win-back offers or lower-priority marketing.

## 13. Business Insights

The dashboard provides a direct view of the size and revenue contribution of each customer segment. The Champions KPI shows that 25.2% of the customer population belongs to the Champions segment. The segment overview can be used to compare customer counts and revenue, while the Recency–Monetary view helps identify differences between recent activity and customer value.

## 14. Business Recommendations

Focus retention and loyalty activities on Champions. Use personalized re-engagement campaigns for At Risk customers. Encourage Price-Sensitive / Low-Value customers to increase basket size through bundles and cross-selling. Use targeted win-back offers for Dormant / Lost customers. Recalculate customer segments periodically so that changes in purchasing behaviour are reflected in the dashboard.

## 15. Project Workflow

Customer/Transaction Data → Data Cleaning → Customer-Level Aggregation → RFM Analysis → Feature Preparation → K-Means Clustering → Segment Interpretation → HTML/Plotly Dashboard → Business Insights → Recommendations.

## 16. Conclusion

The Customer Segmentation Dashboard transforms transaction data into meaningful customer groups using RFM analysis and K-Means clustering. The project demonstrates practical skills in Python data analysis, customer-level feature engineering, machine learning, visualization, and dashboard development. The final dashboard makes customer behaviour easier to understand and supports targeted retention and marketing decisions.